Adaptive Coherent Multilook GLRT for SAR Tomography Detection

Nabil Haddad, Karima Hadj-Rabah, *Member, IEEE*, Gilda Schirinzi, *Senior Member, IEEE*, and Azzedine Bouaraba

***Abstract*—In recent years, generalized likelihood ratio test (GLRT) scatterers' detection in the context of synthetic aperture radar tomography (TomoSAR) has gained great interest from the remote sensing scientific community. This is due to its effectiveness in identifying scatterers within each single azimuth–range resolution cell, particularly in urban areas. The multilook GLRT (M-GLRT) variant offers more satisfactory results at the expense of spatial resolution deterioration, by jointly exploiting the neighboring information of the pixel to be reconstructed. In this context, coherent and incoherent formulations can be adopted. The former provides better performance in the assumption of constant reflectivity of the pixels in the considered neighborhood, while the latter is much more robust with respect to the violation of this assumption. In this article, an adaptive formulation of the coherent and incoherent GLRT is presented with the aim of improving scatterers' detection and their height estimation. The method is based on an adaptive window setting, to select adjacent pixels with similar height and reflectivity characteristics. A detailed study and analysis of the proposed adaptive coherent M-GLRT (ACM-GLRT) detector has been conducted and validated through simulations along with comparison to both standard and adaptive formulations of incoherent M-GLRT. Experimental findings from a real dataset acquired by the German TerraSAR-X (TSX) over the city of Naples (Italy) demonstrate the performance improvement of our proposed approach.**

***Index Terms*—Adaptive context, coherent detection, generalized likelihood ratio test (GLRT), height map, synthetic aperture radar tomography (TomoSAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) techniques have been used in many disciplines related to Earth and environmental sciences, due to various parameters and properties that can be extracted from radar data [1]. Among those information is the height profile of the ground scene, which can be estimated via SAR tomography (TomoSAR) [2], [3]. As an advanced 3-D technique, TomoSAR's main objective is retrieving the vertical distribution of the scatterers inside each azimuth–range resolution cell. This is possible by means of several acquisitions coherently combined using the interferometric configuration, to establish a linear inverse problem to be solved in the direction perpendicular to the azimuth–range plane [4].

In practical situations, TomoSAR sensing system faces a number of limitations mostly related to the tracks' constellation. Nonuniformly distributed acquisitions along the baseline axis and their reduced number introduce artifacts such as increased height ambiguity, decreased vertical resolution, and the presence of unwanted sidelobes [5]. Another factor to be considered is the noise addition [6], whose sources can be classified into: 1) geometric distortions inherent to the acquisition system and the surface topography and 2) the different decorrelation issues due to the multibaseline multitemporal configuration. Those artifacts can be addressed through either a preprocessing denoising [6], [7] or an advanced inversion method whose concept takes into account the noise issue such as subspace-based methods [8], [9], [10], statistical regularization techniques [11], [12], [13], and compressive sensing framework [14], [15], [16].

TomoSAR problem can also be approached differently. In fact, De Maio et al. [17] were the first to formulate it as a detection problem by fixing the false alarm rate and then considering two hypotheses: the absence or the presence of a single scatterer within the azimuth–range resolution cell. They concluded that the proposed generalized likelihood ratio test (GLRT) provides a better outcome compared with both Rao and Wald tests. Pauciullo et al. [18] extended the GLRT test into a double-scatterer case scenario through two modifications. The first one analyzes the amount of useful data energy in the signal subspace without cancellation, while the second one incorporates the cancellation stage into this sequential process. These detectors are limited by the Rayleigh resolution.

To offer super-resolution capabilities, Budillon and Schirinzi [19] introduced the support GLRT variant. Good performances were achieved at the expense of massive high computational time. De facto, the latter is due to the required multidimensional search that increases with a combinatorial law blue added to the grid size and density. To address this issue, fast support GLRT has been proposed in [20]. It maintains satisfactory vertical resolution with an acceptable processing cost at the expense of a slight decrease in the height accuracy of the detected scatterer. To improve the detection performance

Received 11 August 2024; revised 25 February 2025 and 23 June 2025; accepted 21 July 2025. Date of publication 25 July 2025; date of current version 5 August 2025. *(Corresponding author: Karima Hadj-Rabah.)*

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Digital Object Identifier 10.1109/TGRS.2025.3592847

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further, Luo et al. [21] used support GLRT as a postprocessing where they performed a reconstruction using the compressive sensing method to eliminate the less significant elevation positions for the search grid of GLRT detector.

Common advantage of the aforementioned methods is the preservation of spatial resolution from being single-look detectors. However, they suffer from the risk of missing scatterers in regions with a low signal-to-noise ratio (SNR). To solve this matter, the multilooking variant can be of great use. For that, Pauciullo et al. [22] extended the original version of GLRT proposed in [17] to the multilooking detection of single and double scatterers. As a means to attain super-resolution proficiency while maintaining a constant false alarm, the work in [23] combined Capon estimator with multilook fast support GLRT.

An efficient averaging operation requires the selection of surrounding resolution cells with similar backscattering properties. In all the above detectors, the multilooking has been performed by the classical box-car window. Unfortunately, the latter does not satisfy the previous fundamental requirement. It considers the immediate neighbors and gives them the same weight. As an alternative, an adaptive nonlocal approach can be used through the search of similar patches in a larger window defined as a searching window, as well as an efficient estimate of the weights from the data itself [24]. These methods automatically determine the windows of statistically homogenous pixels using different tests such as Kolmogorov–Smirnov [25], [26].

The nonlocal-based technique can be considered an effective framework, as it allows for a more flexible selection of samples. Although its sufficiency depends on the similarity criterion used to quantify pixel homogeneity, which becomes increasingly challenging when a large number of acquisitions are available for time-series analysis [27].

In the context of GLRT, Verde et al. [28] compared the performance of component extraction and selection SAR (CAESAR), developed to optimize multilook processing in multitemporal and multibaseline SAR techniques, with multilook GLRT (M-GLRT) for permanent scatterer detection. While the M-GLRT demonstrated superior detection performance under the assumed statistical model, the authors highlighted several practical advantages of CAESAR. One important advantage is its constant false alarm probability (PFA), which remains independent of the number of looks. This property is particularly useful in mixed urban and nonurban environments, where it simplifies the implementation of adaptive detection schemes with varying multilook factors. In contrast, the M-GLRT requires spatially varying thresholds to maintain a constant PFA across regions with differing numbers of looks, adding complexity to its implementation. Furthermore, it is more sensitive to deviations from the assumed phase model, leading to a higher likelihood of false positives in cases where phase disturbances are present. However, despite these challenges, the M-GLRT exhibits notable robustness against model mismatch, particularly in high-SNR scenarios, making it a valuable benchmark for evaluating detection schemes in heterogeneous environments.

A recent work by Dănişor et al. [29] further underscores these points through a detailed performance comparison of CAESAR, SqueeSAR, and multilook GLRT. The conducted analysis highlighted on one hand the outperformance of SqueeSAR which relies on phase information over CAESAR in areas with high-amplitude inhomogeneity. On the other hand, it pointed out that CAESAR generally achieves detection curves closer to the multilook GLRT considered as a reference.

In this article, our study focuses on enhancing the detection's final fulfillment regarding targets' discrimination and noise effect mitigation. For that purpose, our contribution consists of a different design and implementation of an M-GLRT-based detector. First, coherent integration is included in the maximum likelihood minimization process with the aim of exploring both the components of each multibaseline image within the stack. Second, an adaptive cell neighbors' selection is introduced to adaptively select the best neighborhood in which the scatterers' reflectivity and elevation can be assumed constant with an acceptable approximation. By doing so, critical scene details can be preserved and coherent integration can be carried out. Finally, the application of the proposed variant of the GLRT detector on simulated and real datasets is evaluated through the use of several qualitative and quantitative measurements.

The rest of this article is organized as follows. Section II presents a brief review of the tomographic SAR signal model. Section III is devoted to a detailed description of the GLRT detector with emphasis on our proposed modification. The experimental findings are exposed and discussed in Section IV. Finally, a brief conclusion along with some perspectives is presented in Section V.

II. Data Model

TomoSAR process achieves 3-D reconstruction of the observed scene by gathering M single-look complex images with slightly different view angles, as shown in Fig. 1. First, the multibaseline preprocessing step should be accomplished, including multiple registrations of $(M-1)$ images in accordance with the same reference image, phase calibration through deramping [2], atmospheric correction, and flat Earth phase compensation [3]. At this stage, the mth interferometric image in a fixed pixel u_m can be expressed as the coherent summation of the complex reflectivities of all the scatterers located within the corresponding azimuth–range resolution cell. The different contributions are superimposed to the clutter-plus-noise signal w_m, representing components from noise and incoherent scatterers

$$u_m = \int_{\Delta s} \gamma\left(s'\right) \exp\left(-j4\pi \frac{b_{\perp m}}{\lambda r_0} s'\right) ds' + w_m \tag{1}$$

where $b_{\perp m}$ is the perpendicular baseline, λ is the operating wavelength, r_0 is the slant-range distance between the SAR antenna and the ground pixel, s' is the elevation direction orthogonal to range and azimuth, and Δs is the ground scene extension along the elevation direction.

In the case of urban applications, the number of coherent scatterers present within each resolution cell is limited [19]. In this article, we assume the existence of at most a single dominant target for each azimuth–range resolution cell located at the unknown elevation s so that $\gamma(s') = \gamma\delta(s' - s)$,

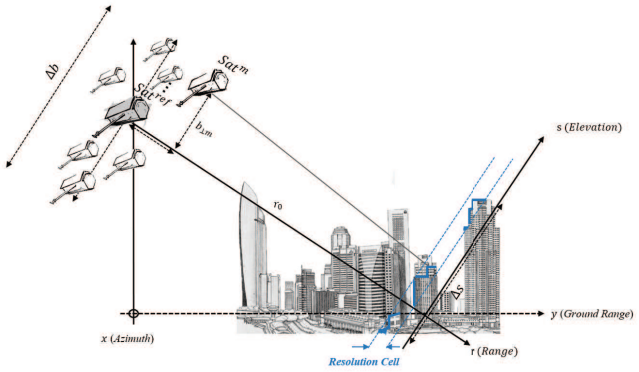


Fig. 1. Geometry of an SAR tomographic system.

where $\delta(\cdot)$ denotes the Dirac delta function. This assumption facilitates addressing the tomographic reconstruction problem by introducing the following two statistical hypotheses:

$$H_0 : u_m = w_m \tag{2}$$

$$H_1 : u_m = \gamma \exp\left(-j4\pi \frac{b_{\perp m}}{\lambda r_0} s\right) + w_m, \quad m = 1, \ldots, M \tag{3}$$

where H_0 and H_1 are the hypotheses modeling the cases describing the absence and presence of a coherent scatterer, while γ and s denote the deterministic complex reflectivity and the elevation of the dominant scatterer, respectively. The expanded form in (2) and (3) can be converted into the subsequent compact vector representation

$$H_0 : \boldsymbol{u} = \boldsymbol{w} \tag{4}$$

$$H_1 : \boldsymbol{u} = \boldsymbol{\varphi}(s)\gamma + \boldsymbol{w} \tag{5}$$

where $\boldsymbol{u}$ and $\boldsymbol{w}$ are the representative vectors of observation data and noise samples, respectively, while $\boldsymbol{\varphi}$ accounts for the steering vector that can be written as

$$\boldsymbol{\varphi}(s) = \left[\exp\left(-j4\pi \frac{b_{\perp 1}}{\lambda r_0} s\right), \ldots, \exp\left(-j4\pi \frac{b_{\perp M}}{\lambda r_0} s\right)\right]^{\mathrm{T}} \tag{6}$$

where ^{T} denotes the transpose.

III. GLRT Detector

In this section, we formulate the detection problem within the GLRT framework, following both a single-look and a multilook approach, and we introduce our proposed improved variant.

A. Single and Multilook Formulation

The basic principle of GLRT detector consists of solving simultaneously two problems: detection and localization. For that, the primary assumption is that the clutter-plus-noise signal affecting multibaseline data is a white random process following a complex circularly Gaussian distribution [30] with zero mean and covariance matrix $\sigma^2 \boldsymbol{I}_M$ ($\boldsymbol{I}_M$ denotes the $M \times M$ identity matrix). Assuming the reflectivity to be deterministic, the data vector $\boldsymbol{u}$ is still Gaussian with covariance matrix $\sigma^2 \boldsymbol{I}_M$ and mean value depending on two case scenarios: under the hypothesis H_0 where it is equal to zero, and under the hypothesis H_1 where it becomes equal to $\boldsymbol{\varphi}(s)\gamma$. Therefore, the probability density function (pdf) of the data vector under these two hypotheses can be written as

$$f_{H_0}\left(\boldsymbol{u}; \sigma^2\right) = \frac{1}{\pi^M \sigma^{2M}} \exp\left(-\frac{\boldsymbol{u}^H \boldsymbol{u}}{\sigma^2}\right) \tag{7}$$

$$f_{H_1}\left(\boldsymbol{u}; \gamma, s, \sigma^2\right) = \frac{1}{\pi^M \sigma^{2M}} \exp\left(-\frac{(\boldsymbol{u} - \boldsymbol{\varphi}(s)\gamma)^H (\boldsymbol{u} - \boldsymbol{\varphi}(s)\gamma)}{\sigma^2}\right) \tag{8}$$

with $()^H$ being the Hermitian operator. The GLRT is expressed with the subsequent detection rule that depends on the likelihood function of $\boldsymbol{u}$ as follows:

$$\Lambda_0 = \frac{\max\limits_{\gamma, s, \sigma^2} f_{H_1}\left(\boldsymbol{u}; \gamma, s, \sigma^2\right)}{\max\limits_{\sigma^2} f_{H_0}\left(\boldsymbol{u}; \sigma^2\right)} \underset{H_0}{\overset{H_1}{\gtrless}} \eta_0. \tag{9}$$

The expression of the estimates of γ and σ^2 maximizing numerator and denominator of (9) can be found in a closed form and substituted in (9), so that, after some mathematical manipulations, a simplified form of the test can be obtained [17]

$$\Lambda_{sl} = \frac{\max\limits_{s} \frac{1}{M} |\boldsymbol{\varphi}(s)^H \boldsymbol{u}|^2}{\boldsymbol{u}^H \boldsymbol{u}} \underset{H_0}{\overset{H_1}{\gtrless}} \eta_{sl}. \tag{10}$$

The threshold η_{sl} can be numerically estimated using Monte Carlo simulation for a fixed value of the probability of false alarm (Pfa), which is the conditional probability $P(H_1/H_0)$.

Expression (10) represents the single-look GLRT test, since the scatterer detection and localization are performed using only the multibaseline data vector without any spatial averaging operation involved. The probability of the scatterer detection (for a fixed Pfa) and the estimation accuracy of its elevation, achieved using the single-look test (9), strongly depend on the scatterer coherence among the M acquisitions blue and the SNR. When M is low and/or the scatterer is scarcely coherent, the test performance can be improved by considering a cluster of $L-1$ pixels in a local neighborhood surrounding the pixel of interest, under the pixel similarity presumption, so that the number of data is increased by a factor of L.

Two hypotheses can be considered according to [22] regarding the targets on the ground. In fact, they can be classified, based on their fluctuation characteristics over time, into two groups: fluctuating and nonfluctuating scatterers. In this way, an M-GLRT can be derived as follows.

B. Incoherent M-GLRT

We consider a neighbor of $L-1$ pixels surrounding the pixel of interest and having all the targets responding with different reflectivities γ_i yet located all at the same elevation s (note that $i = 1, \ldots, L$). The noise samples are assumed statistically independent among the L pixels and their variance σ^2 is considered constant. In this case, we have LM data and $L+2$ unknowns (L reflectivity values γ_i, the elevation s, and the noise variance σ^2). In this hypothesis, the GLRT can be written as follows [22]:

$$\Lambda_{ml} = \frac{\max\limits_{s} \sum_{l=1}^{L} \frac{1}{M} |\boldsymbol{\varphi}(s)^H \boldsymbol{u}_l|^2}{\sum_{l=1}^{L} \boldsymbol{u}_l^H \boldsymbol{u}_l} \underset{H_0}{\overset{H_1}{\gtrless}} \eta_{ml} \tag{11}$$

where the threshold η_{ml} is also estimated using Monte Carlo simulation by setting the Pfa to a predetermined value.

This formulation of M-GLRT involves incoherent integration over the looks [see (11)], since only the magnitude information of the signal reconstructed from each look is being averaged, while its phase is discarded (incoherent averaging). The adoption of an incoherent model along the looks derives from the assumption that the targets have different reflectivities from look to look within the considered neighborhood [31].

The strength of incoherent GLRT resides in its robustness against reflectivity variations and phase uncertainties. However, its drawback mainly consists of being limited in terms of integration gain.

C. Coherent M-GLRT

We also consider the nonfluctuating case in which the targets are assumed to be stable in terms of backscattering reflectivity within different looks. From a signal processing point of view, it is known that considering a nonfluctuating scenario allows to perform a coherent integration over multiple looks, which improves the SNR level and, consequently, improves the detection performance. Hence, considering a nonfluctuating scenario for both acquisitions and looks is a worthy-to-try hypothesis.

When an L-pixel neighborhood of the observed pixel with scatterers of constant reflectivity γ and located at a constant elevation s is available, the number of the unknown parameters decreases to 3 (the constant reflectivity γ, elevation s, and noise variance σ^2), while the number of data is still LM. In this assumption, the M-GLRT performance can be improved, both from the detection and accurate elevation estimation viewpoint, since the ratio between the number of data and the number of unknowns is increased. Hence, the formulation of the M-GLRT test can be expressed as follows:

$$\Lambda'_{ml} = \frac{\max_s \frac{1}{LM} \left| \sum_{l=1}^{L} \boldsymbol{\varphi}(s)^H \boldsymbol{u}_l \right|^2}{\sum_{l=1}^{L} \boldsymbol{u}_l^H \boldsymbol{u}_l} \underset{H_0}{\overset{H_1}{\gtrless}} \eta'_{ml}.X \tag{12}$$

This multilook test considers the coherent average of the complex phase compensated looks $\boldsymbol{\varphi}(s)\boldsymbol{u}_l$. Consequently, the test is expected to be better performing when the adopted assumptions are satisfied with a good approximation. Note that the full mathematical demonstration of (12) is available in the Appendix. The benefits of Coherent GLRT can be summarized in three points as follows.

1) *Improved Detection Sensitivity*: by coherently averaging the phase compensated looks $\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l$, the detection probability for weak targets masked by noise can be enhanced.
2) *Higher Integration Gain*: coherent integration of looks allows a higher integration gain compared with the incoherent GLRT, since both phase and amplitude are involved.
3) *Increased SNR Level*: due to coherent integration gain, which is proportional to the considered number of looks, a higher SNR can be obtained.

Despite the advantages offered by coherent integration, it presents significant challenges that must be carefully addressed in the detector's design and implementation. A primary consideration involves the assumption of nonfluctuating targets across space. To fully exploit the benefits of coherent integration, the correlation properties of the backscattering across the looks must be preserved. This, however, is not always feasible, especially in urban environments where backscattering is highly heterogeneous. In such scenarios, the level of multilooking is generally limited or entirely absent, which can significantly degrade performance by reducing coherence and increasing noise levels.

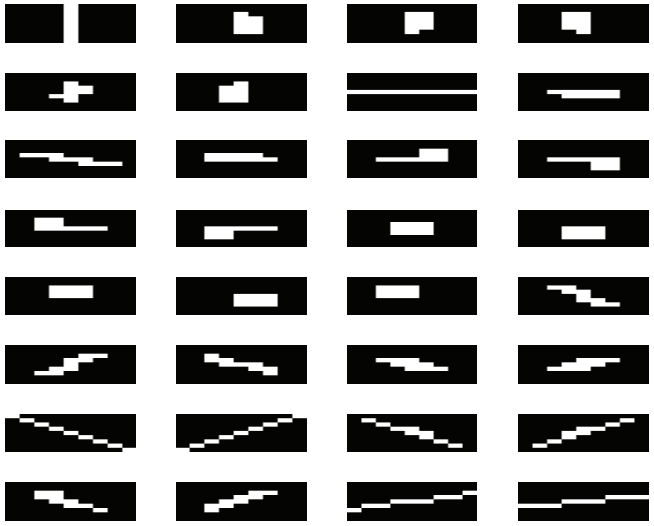


Fig. 2. Considered patches for adaptive neighborhood selection.

One solution involves carefully selecting pixels with similar backscattering properties from the neighborhood surrounding the pixel under observation. However, traditional boxcar windows fail to account for local variations within the selected samples, as they rely solely on spatial proximity without incorporating measures of similarity. Despite the criterion considered in nonlocal windows [27], [32] that attempts to group patches based on statistical similarity, the latter does not guarantee comparable reflectivity distributions, which are crucial for coherent integration accuracy.

In this article, we propose a local neighborhood selection adapted to urban scenes where very regular structures with sharp edges are typically present in the images. In particular, we consider the choice of one of N_w neighborhoods of L pixels, located within a scan $L \times L$ window. In this way, a set of directional neighborhoods are created. The neighborhood selection criterion is based on the maximization of the likelihood ratio (11) and (12). This approach is similar to the one followed in [33] for adaptively filtering noise along interferogram fringes using directionally dependent windows.

The binary masks identifying a set of $N_w = 32$ directional neighborhoods of $L = 9$ pixels, fitting well the structures' forms and allowing better adaptation to scene heterogeneity, are shown in Fig. 2. The generation of other sets of neighborhoods can be easily done. The principle is that the choice of a proper directional neighborhood, adaptively matching the scene's varying characteristics, allows the requirement for scatterers of constant reflectivity and height to be met more easily.

TABLE I
TSX SYSTEM PARAMETERS

Parameter	Quantity
wavelength	0.0311 [m]
Incidence angle	28.75°
Range Distance	579.4 [km]
Number of images	27
Total Baseline	751.6 [m]

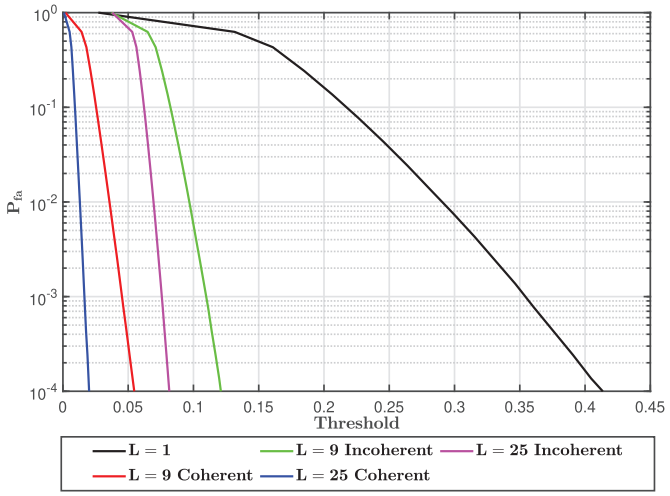


Fig. 3. Pfa versus the detection threshold for different numbers of looks in both coherent and incoherent integration scenarios.

To formulate the adaptive selection for the coherent M-GLRT considering N_w different masks, and letting Λ'_{mli} be the GLRT value computed for the ith mask. The adaptive likelihood ratio can be given by

$$\Lambda'_{\max} = \max\left(\Lambda'_{ml1}, \Lambda'_{ml2}, \Lambda'_{ml3}, \ldots, \Lambda'_{mlN_w}\right) \tag{13}$$

where $\Lambda'_{\max}$ represents the maximum GLRT value among all the computed GLRT values from the predefined masks.

IV. EXPERIMENTAL RESULTS

In this section, we investigate the performance and robustness of the proposed adaptive coherent M-GLRT (ACM-GLRT) detector using both the simulated tomographic data according to TerraSAR-X (TSX) satellite parameters and real SAR images acquired by the same sensor.

A. Simulated Results

In the first place, controlled environments are considered according to the system parameters reported in Table I. In these experiments, we take into account a neighborhood with $L = 9$ and $L = 25$ looks for the multilooking framework, to examine the effectiveness of coherent GLRT compared with incoherent GLRT. The analysis is conducted in terms of false alarm and detection (Pd) probabilities, along with the root mean square error (RMSE), which are assessed using Monte Carlo simulation over 10^6 realizations of data. Our study can be split into two parts as follows.

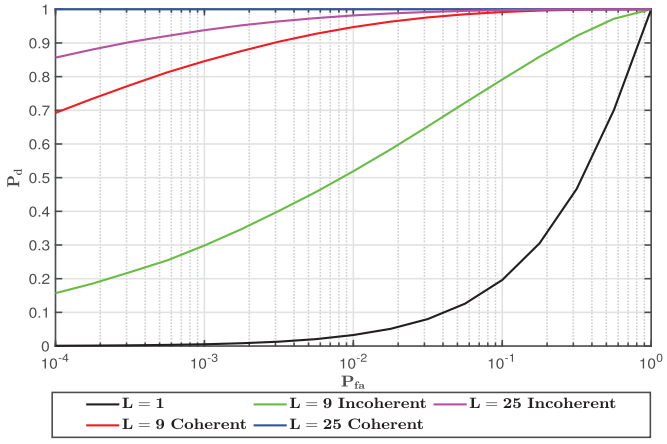


Fig. 4. Receiver operating characteristic curves for an SNR of −15 dB, comparing detection performance of different GLRT methods versus false alarm rate (P_{fa}).

1) Effectiveness of Coherent Integration Within GLRT Framework: The behavior of Pfa as a function of the threshold can be analyzed from the graph of Fig. 3, for different numbers of looks $L = 1, 9, 25$. As expected for all the cases, the Pfa values decrease with the increase in the threshold values independently of the detection technique and the number of looks. In addition, for a fixed Pfa, the threshold value decreases as the numbers of looks L increase. A similar behavior can also be perceived for a fixed number of looks where the coherent integration has a threshold value smaller than the incoherent one.

Fig. 4 illustrates the GLRT receiving operating characteristics (ROC) curves for the probability of detecting one scatterer obtained under distinct P_{fa} values, considering single-look along with multilook scenarios ($L = 9$ and $L = 25$) for a fixed SNR value of −15 dB. We can perceive from the displayed plots that the use of the coherent GLRT gives a better detection performance compared with the incoherent GLRT when the number of looks is equal to 9. These findings can be further improved for a higher number of looks, as shown by the $L = 25$ graphs. The latter outcome demonstrates superior performance of the coherent GLRT.

This improvement in detection performance arises because the coherent GLRT effectively exploits phase information across multiple looks, enabling it to achieve higher sensitivity. However, while increasing the number of looks enhances detection capabilities, maintaining spatial coherence becomes increasingly challenging as the multilook window expands, especially in scenarios with significant scene heterogeneity.

The PDFs of the two tests (11) and (12) carried out for $L = 9$ and $L = 25$ are displayed in Figs. 5 and 6, with the aim of showing the ability of the coherent GLRT and incoherent GLRT to differentiate between the different statistical hypotheses H_0 and H_1. It can be discerned that the PDFs in the two hypotheses are overlapped for both the detectors in the case of nine looks, but the overlapping area is notably smaller for the coherent GLRT [see Fig. 5(b)], thus denoting its better discrimination capacity respect to the incoherent one.

For a number of looks $L = 25$, we can observe a slight overlap between the coherent GLRT PDFs under H_0 and

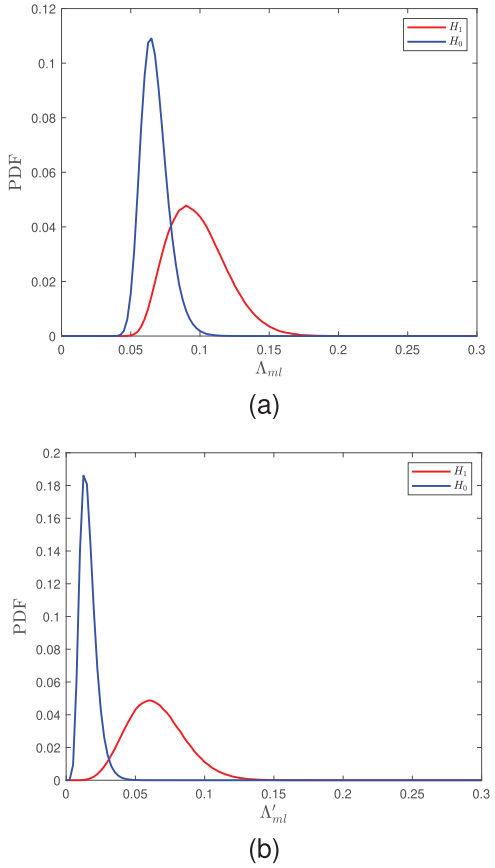


Fig. 5. PDFs for hypotheses H_0 (null hypothesis) and H_1 (alternative hypothesis) under (a) incoherent GLRT and (b) coherent GLRT, computed for $L = 9$, as a showcase for the separability extent of the two hypotheses.

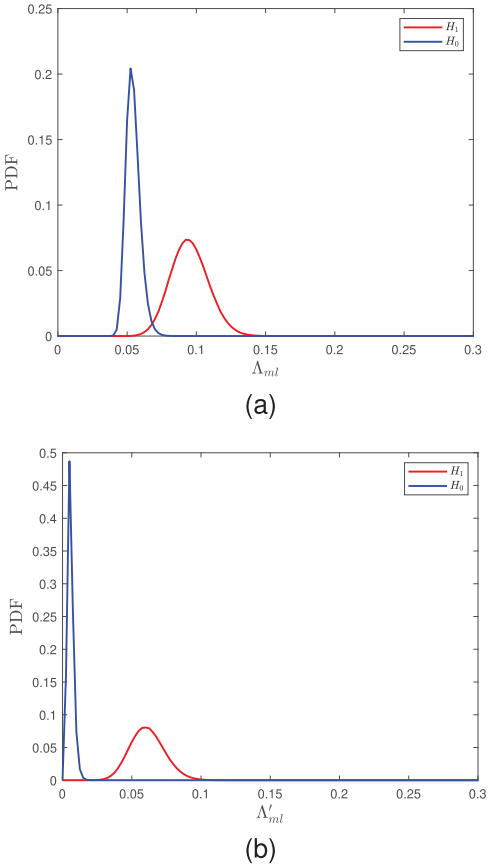


Fig. 6. PDFs for hypotheses H_0 (null hypothesis) and H_1 (alternative hypothesis) under (a) incoherent GLRT and (b) coherent GLRT, computed for $L = 25$, as a showcase for the separability extent of the two hypotheses.

TABLE II
DEFLECTION COEFFICIENT PARAMETER

GLRT Detector	Number of looks	Deflection coefficient
Incoherent	9	10.0684
Coherent	9	67.2378
Incoherent	25	64.2548
Coherent	25	649.8119

H_1 [see Fig. 6(b)]. This indicates that the coherent test can effectively distinguish between the two hypotheses. These attainments can be further investigated and confirmed by computing the deflection coefficient given by [34]

$$D = \frac{(m_0 - m_1)^2}{V_0} \tag{14}$$

where m_0 and V_0 account correspondingly for the mean and variance of the pdf under hypothesis H_0, while m_1 denotes the mean of H_1 pdf. The higher the deflection coefficient, the easier it is to distinguish between H_0 and H_1 hypotheses, leading to a higher detection performance [34].

The values of the deflection coefficient for both the coherent and incoherent detectors can be found in Table II for both the numbers of looks considered ($L = 9$ and $L = 25$). It can be observed that the deflection coefficient of the coherent detector surpasses that of the incoherent detector, indicating that the coherent test offers greater capability to differentiate between the two hypotheses H_0 and H_1. This outcome highlights the result obtained previously.

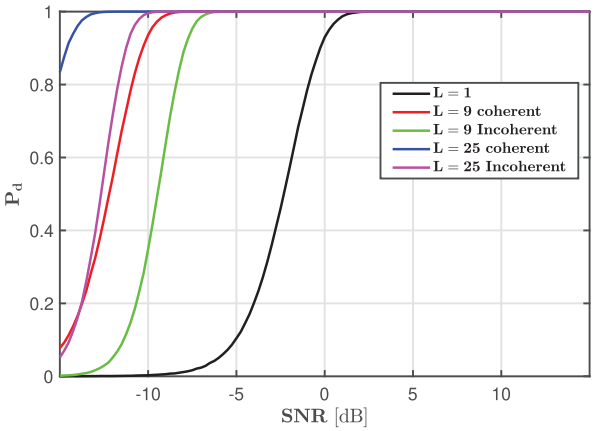


Fig. 7. Detection probability as a function of SNR for a fixed value of Pfa (10^{-4}).

Moreover, to reveal the effectiveness of the detectors with respect to noise resilience and elevation estimation precision, Fig. 7 reports the detection probability as a function of SNR for a fixed value of Pfa (10^{-4}). We note that the Pd increases with the increase in SNR value and the number of looks to achieve the unity for a certain value. Nevertheless, it can be observed that the coherent GLRT curves reach the unity

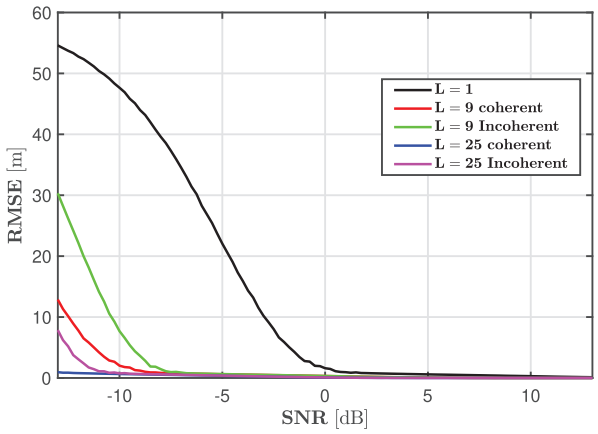


Fig. 8. RMSE as a function of SNR for a fixed value of Pfa (10^{-4}).

value at a lower SNR compared with the incoherent GLRT for similar number of looks. Analogously, Fig. 8 illustrates the variation in the retrieved heights RMSE as a function of the SNR values. It can be seen that the RMSE decreases when the SNR value and the number of looks increase, with a remarkable superiority of the coherent GLRT over the incoherent GLRT.

The analysis conducted so far demonstrates the superiority of coherent integration over incoherent integration within the framework of GLRT detection in terms of hypothesis separability and detection along with height estimation.

2) Effectiveness of Adaptive Detection Within GLRT Framework: Performing multilooking on a neighborhood introduces errors due to height and reflectivity variations among scatterers within the neighborhood, which can negatively impact the detection performances. To address these challenges, we proposed the application of an adaptive approach using a set of patches for the neighborhood selection represented in Fig. 2. In fact, aligning the window selection with the geometry of the urban structure is crucial for capturing the desired features effectively.

The simulation for this section considers the scene with objects resembling an urban area represented in Figs. 9(a)–11(a). In this case, a neighborhood covering a 3×3 window ($L = 9$) is used for the two variants of the GLRT (coherent and incoherent) under the nonadaptive approach. Although, for the adaptive approach, we used the masks in Fig. 2, maintaining the same number of looks as the previous nonadaptive approach. The reason for selecting a nine-look window instead of a 25-look one is to make sure that the pixels are located in constant height neighborhoods, while introducing only a slight loss in spatial resolution.

To evaluate the effectiveness of the proposed variant, we generated scatterers located at 30- and 60-m heights under an SNR value equal to -4 dB. Three distinct scenarios have been considered: for the first one, illustrated in Fig. 9 both the scatterers have an equal reflectivity value. The second scenario represented in Fig. 10 involves two different reflectivity values, where the scatterer positioned at 60 m has higher value than the one at 30 m. In contrast to this scenario, for the last one depicted in Fig. 11, the reflectivity of the scatterer located at 30 m is higher than the one positioned at 60 m.

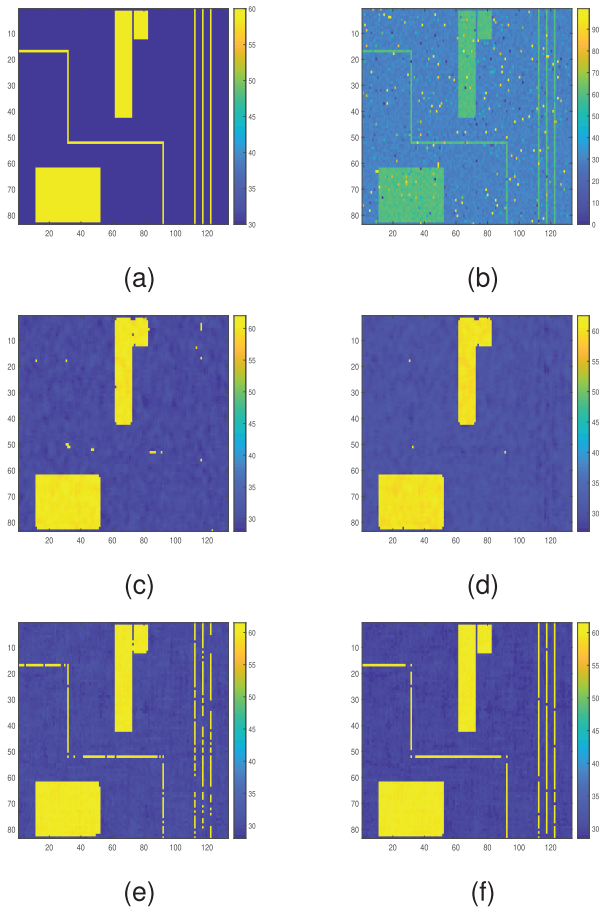


Fig. 9. Simulated scene with two scatterers located at 60 and 30 m having equal reflectivity, for a fixed SNR value of –4 dB. (a) Simulated height map. The obtained height map using (b) single-look GLRT, (c) multilook incoherent GLRT, (d) multilook coherent GLRT, (e) adaptive multilook incoherent GLRT, and (f) adaptive multilook coherent GLRT.

By examining these cases, we aim to assess the performance of the methods under different reflectivity conditions, providing insights into their ability to handle variations in scatterer heights and corresponding reflectivity characteristics. Their attainments should offer valuable information regarding the adaptability and robustness of the methods in realistic scenarios.

Fig. 9 illustrates the height maps obtained using different detectors in the first situation (with nonconstant height and constant reflectivity). It can be observed that the single-look GLRT method [Fig. 9(b)], as expected, effectively preserves spatial resolution reconstructing fine details, especially in thin lines, while it suffers from poor height estimation. On the contrary, the coherent GLRT and incoherent GLRT methods offer more accurate height estimation for thicker structures but lack the ability to capture fine details due to the multilooking effect, which introduces a tradeoff between height reconstruction accuracy and spatial resolution reduction. In contrast, the adaptive methods demonstrate significant improvements in both height estimation and detail preservation, with the ACM-GLRT showing particular superiority. This enhancement can be attributed to the use of adaptive masks that adapt the reconstruction to the geometric characteristics of the structures, enabling precise height estimation by considering pixels with similar heights.

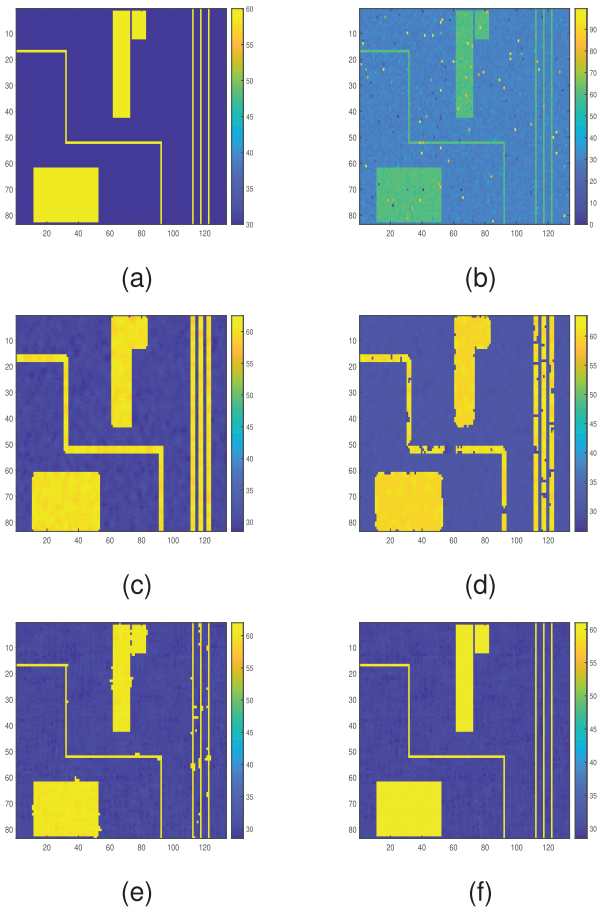


Fig. 10. Simulated scene with a scatterer located at 60 m having higher reflectivity than a second scatterers located at 30 m, for a fixed SNR value of -4 dB. (a) Simulated height map. The obtained height map using (b) single-look GLRT, (c) multilook incoherent GLRT, (d) multilook coherent GLRT, (e) adaptive multilook incoherent GLRT, and (f) adaptive multilook coherent GLRT.

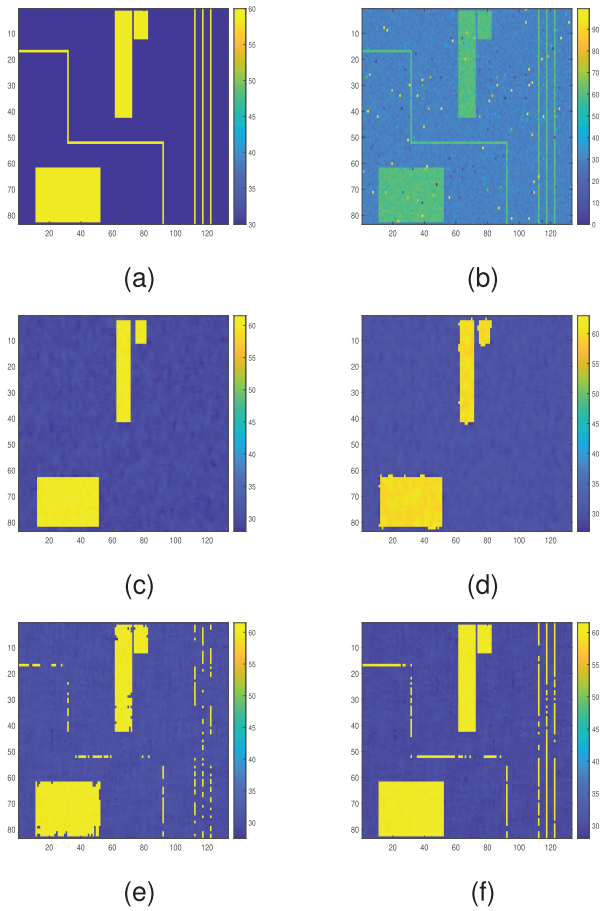


Fig. 11. Simulated scene with a scatterer located at 60 m having lower reflectivity than a second scatterers located at 30 m, for a fixed SNR value of -4 dB. (a) Simulated height map. Obtained height map using (b) single-look GLRT, (c) multilook incoherent GLRT, (d) multilook coherent GLRT, (e) adaptive multilook incoherent GLRT, and (f) adaptive multilook coherent GLRT.

Fig. 10, corresponding to the second context (with nonconstant height and reflectivity), shows that for the single-look GLRT method the outcomes remain unchanged from the previous case, since each pixel is independently estimated unaffected by variations in height or reflectivity of the surrounding pixels. However, in the case of coherent and incoherent M-GLRT, it is obvious that while fine lines are reconstructed, the details appear coarser, leading to a loss of information when two structures are nearby. The latter occurs due to the multilooking process applied to pixels containing different scatterers, causing a bias toward aligning with the scatterer with higher reflectivity. Nevertheless, this issue can be mitigated using adaptive methods, which demonstrate remarkable performance in such scenarios by leveraging their sensitivity to changes in height. By adapting the masks to fit the geometry of the structures, these GLRT variants achieve improved height estimation while preserving fine details.

The findings obtained in the last scenario (with nonconstant height and reflectivity) depicted in Fig. 11 confirm again that the single-look GLRT detector is unaffected by the variations in height or reflectivity. Similarly, the coherent and incoherent GLRT methods achieve accurate height estimation at the cost of losing fine details. This is primarily due to the inclusion of pixels with a scatterer located at 30 m during the multilooking process in neighborhoods containing fine structures with a height of 60 m. The increased reflectivity of the first scatterer dominates height estimation, leading to the loss of fine details. Repeatedly, these drawbacks are mitigated when the adaptive detection approach is used, while the coherent GLRT achieves the best outcomes.

TABLE III
RMSE OF THE SIMULATED HEIGHT MAPS FOR THE THREE CONSIDERED CASE SCENARIOS

Methods	RMSE [m]		
	case1	case2	case3
SL-GLRT	7.2551	7.2898	7.6317
Incoherent GLRT	5.8979	7.1204	8.8023
Coherent GLRT	5.9688	7.1612	9.1961
AIM-GLRT	3.0171	4.8479	2.9159
ACM-GLRT	2.0736	2.5545	0.6453

To validate our latest observation quantitatively, we calculated the RMSE for different GLRT detectors' outcomes across the three considered situations. The computation is laid out in Table III, where the superiority of ACM-GLRT can be spotted. The latter holds the lowest RMSE values across all

the cases, reflecting its adaptability and precision in height estimation. Following ACM-GLRT, the adaptive incoherent M-GLRT (AIM-GLRT) ranks second in performance, offering lower RMSE values than the standard coherent and incoherent GLRT methods.

Among the conventional nonadaptive approaches, the incoherent GLRT method demonstrates better performance than the coherent GLRT which is very sensitive to height change across neighborhood. Yet, these standard methods do not exhibit the same robustness and adaptability as their adaptive peers, particularly in scenarios involving variations in scatterer height and reflectivity.

These results emphasize the advantages of adaptive techniques particularly our ACM-GLRT, which uses adaptive masks tailored to the backscattering and geometric characteristics of the structures. These masks allow for precise height estimation by mitigating issues such as height bias introduced during multilooking processes. As a drawback, adaptive multilooking requires trying different patches to choose the best one that fits the scene geometry, which increases the computational cost.

Consequently, the simulated scenes' attainments show that the ACM-GLRT variant rises as the finest possible compromise in terms of detection performances, height, and reflectivity precision. Hence, in the following, we will refer to it as the proposed detector.

B. Real Data Results

In the second place, real data acquired by the German TSX have been used to validate the conclusions drawn from the simulated results. The dataset's parameters are the same as the ones used for the simulation phase (see Table I). The scene's optical image obtained from Google Earth along with the SAR amplitude map are reported in Fig. 12. It covers an urban region containing man-made buildings of different forms including Piazza del Plebiscito in the Italian city Naples.

Figs. 13 and 14 represent the height maps acquired from a lidar system [Figs. 13(a) and 14(a)] as well as estimated using SAR-based single-look [Figs. 13(b) and 14(b)] and M-GLRT detectors [Figs. 13(c)–(f) and 14(c)–(f)]. In these GLRT-based height maps, the presence of single scatterers is necessarily assumed in the first figure, while only the detected single scatterers having a likelihood ratio greater than the computed threshold is considered for the second one. It can be perceived from all the maps that the multilooking GLRT variants perform better than single-look GLRT since they have less chaotic effects and more homogeneous aspects.

By inspecting Fig. 13(c) and (d), we may argue that in some areas non-AIM-GLRT accomplishes a bit better than nonadaptive coherent GLRT particularly in the flat surface adjacent to the dome. This is most likely due to the strong scattering from the nearby dome structure, causing artifacts or leakage. However, the attainment in Fig. 14(c) and (d) shows that those cells have been discarded by the decision test. In addition, it can be seen from adaptive incoherent and coherent GLRT height maps (Fig. 13(e) and (f), respectively) that the former shows some block-like effect which makes the latter a better option. The first conclusion that can be drawn is that adaptive coherent GLRT performs better than AIM-GLRT, thanks to the coherent superposition which exploits the complete information provided for each single pixel selected in the window.

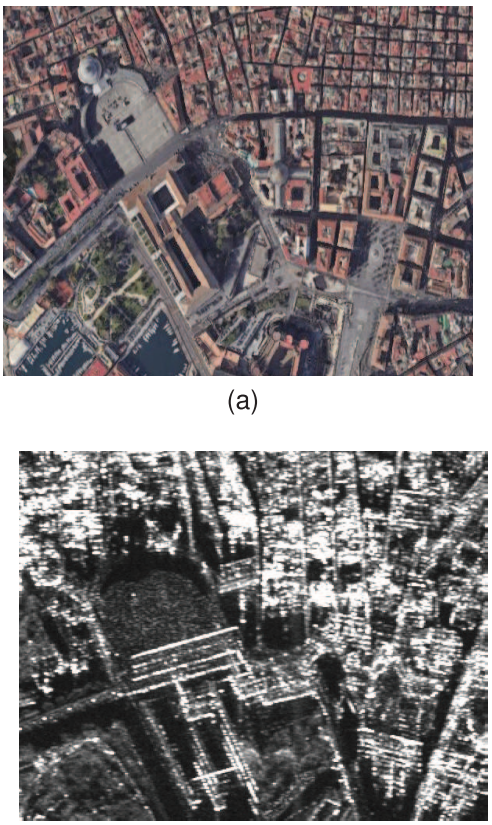


Fig. 12. (a) Google Earth optical image and (b) SAR amplitude map of the scene under study.

The performance of these GLRT variants reveals intricate variations across different scene characteristics. In homogeneous areas with preserved backscattering correlation, the adaptive GLRT methods excel in filtering and scatterer detection. This can be attributed to their capacity to select an adequate neighborhood among the possible ones. Distinctively, the ACM-GLRT adopted as the proposed method demonstrates a subtle yet significant advantage, more evident in complex environments such as the top of San Francesco di Paola Church dome, where it achieves remarkable precision in both detection and height estimation. This improvement is well-expressed in Fig. 14(f), where we can note that most resolution cells' likelihood ratio exceeded the threshold compared with the other cases [see Fig. 14(b)–(e)] since their height maps have important gaps. Hence, it can be concluded that adaptive coherent GLRT provides the most satisfying outcome compared with the lidar height map in terms of scatterers' density, noise reduction, and resolution preservation. Despite these strengths, the method presents some issues which have to be properly managed. In particular, in areas suffering from high decorrelation effects, like the lower left part of the scene, a proper strategy for the GLRT threshold choice, based on

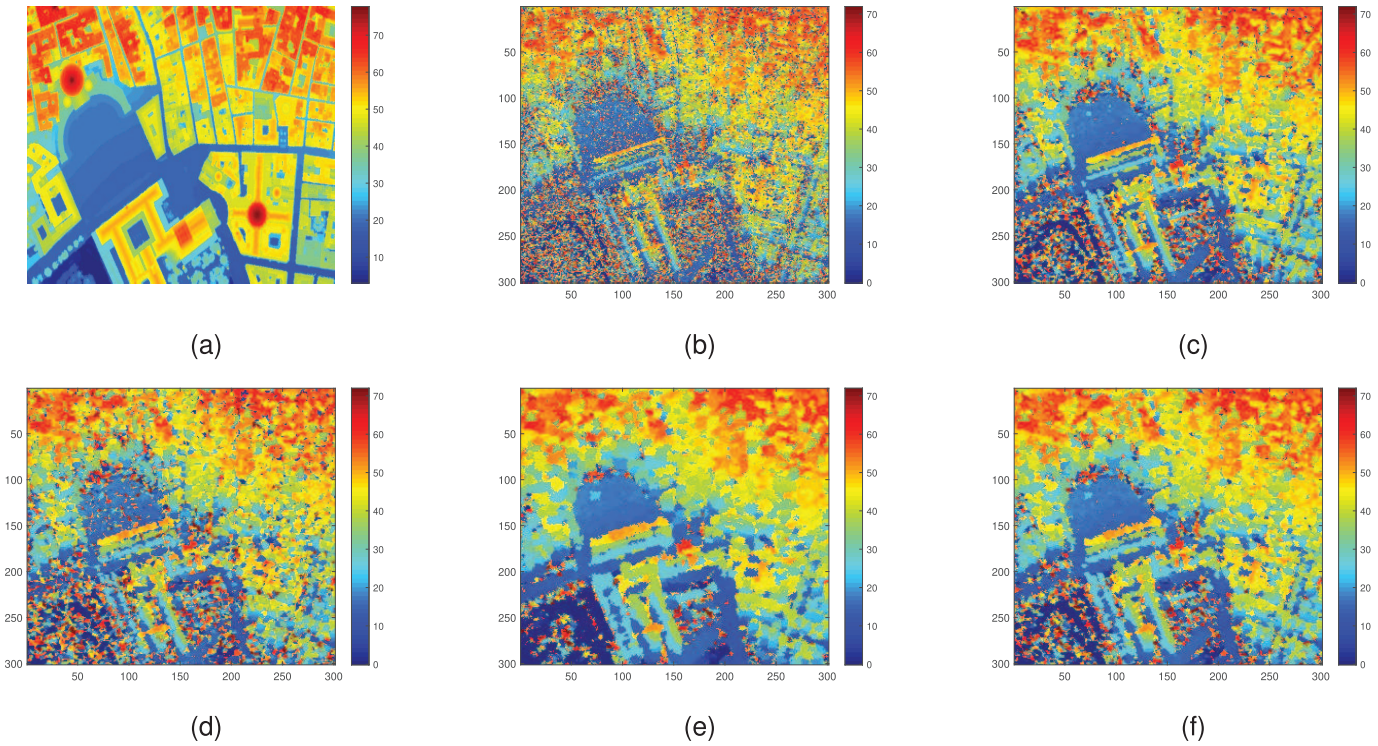


Fig. 13. (a) Lidar height map. The obtained height map of the main scatterers using (b) single-look GLRT, (c) multilook incoherent GLRT, (d) multilook coherent GLRT, (e) adaptive multilook incoherent GLRT, and (f) adaptive multilook coherent GLRT. Note that no thresholding has been applied.

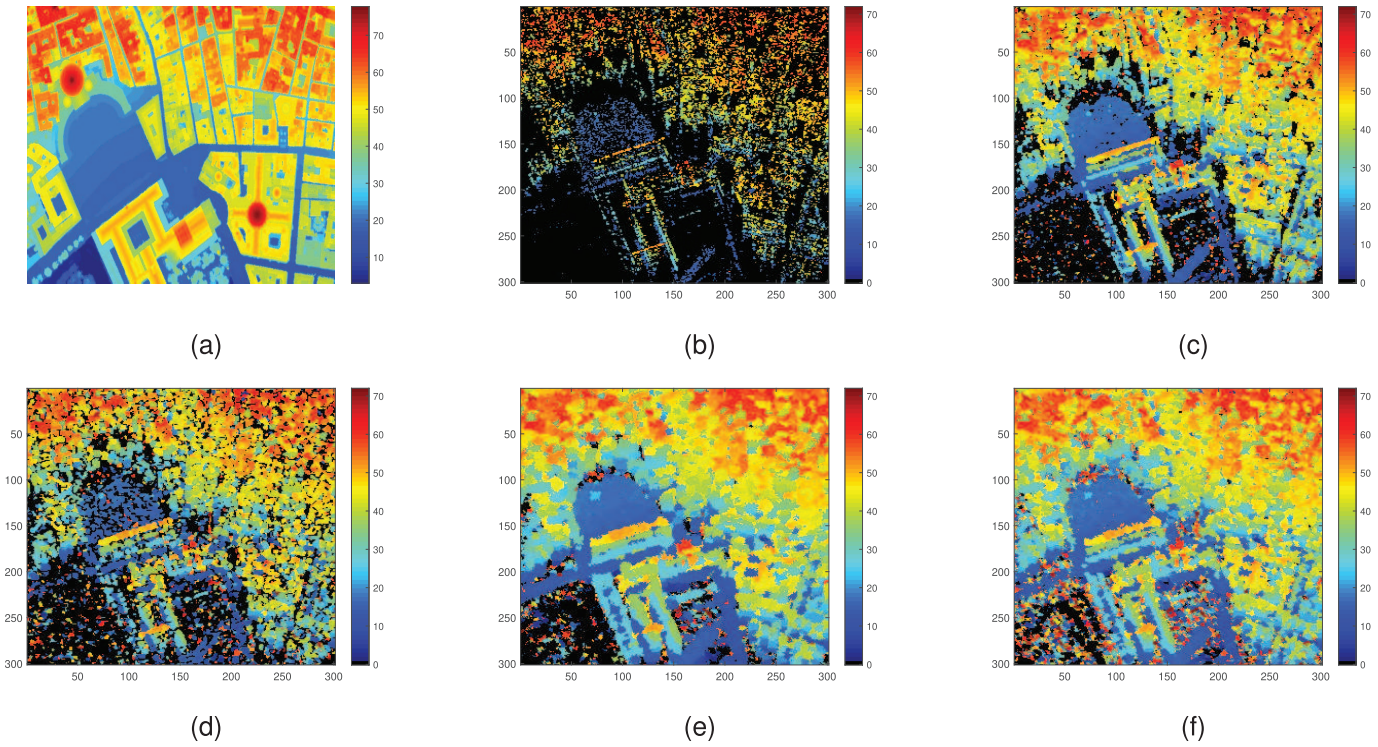


Fig. 14. (a) Lidar height map. The obtained height map of the detected scatterers for a Pfa equal to 10^{-3} using (b) single-look GLRT, (c) multilook incoherent GLRT, (d) multilook coherent GLRT, (e) adaptive multilook incoherent GLRT, and (f) adaptive multilook coherent GLRT.

the equivalent number of looks and average coherence values, should be introduced to filter out pixels that give rise to a noisy height estimation, similar to the approach introduced in [28].

V. Conclusion

In this article, we proposed a variant of the GLRT detector in the context of urban TomoSAR based on two main contributions. The first one consists of an adaptation of a

coherent multilook integration into the GLRT decision test which accumulates the different looks coherently by incorporating the phase information. The second step involves the optimal choice of a directional mask to be used for selecting the best neighborhood to be used in the coherent M-GLRT, on the basis of the statistical properties of the data within the neighborhood. By comparing the GLRT values across masks, the one producing the highest likelihood ratio can be identified. The selected mask represents the most suitable representation of the desired features of the scene, leading to an accurate reconstruction. The proposed ACM-GLRT was developed and its performance was assessed using both the simulated and real data. Compared with the case of traditional incoherent GLRT, adaptive coherent integration proved to be effective in terms of detection sensitivity and increased SNR level. Adaptive detection achieved a better outcome compared with classical nonadaptive case scenarios regarding height precision. The combination of adaptive and coherent detection proved to be the finest adaptation of GLRT detectors into realistic structures' outlines.

Other adaptive GLRT methods have already been introduced, such as the one based on the estimation of local plane parameters [31], which is based on the local representation of the imaged surface by means of planes, whose slopes are adaptively estimated. This method has proved to be able to improve the results when the imaged surfaces on the ground notably deviate from horizontal surfaces, so that the assumption of constant height in the multilook neighborhood is violated. The method takes into account the steering vectors variations in the multilook window by adaptively estimating the parameters of a locally planar surface. The adaptive scheme introduced by the proposed ACM-GLRT, instead, is adaptively searching the neighborhood where both reflectivity and steering vectors (i.e., elevations) can be considered constant. In other words, it allows to mitigate the changes in reflectivity and elevation within the multilook window by selecting the neighborhood with the minor changes. Future work could explore integrating the two adaptive approaches to further improve height estimation and detection performance.

APPENDIX

Under the null hypothesis H_0, the pdf of the observed data $\boldsymbol{u}_l$ for L independent looks is expressed as

$$f_{H_0}\left(\{\boldsymbol{u}_l\}_{l=1}^{L};\sigma^2\right)=\prod_{l=1}^{L}\frac{1}{\pi^M\sigma^{2M}}\exp\left(-\frac{1}{\sigma^2}\|\boldsymbol{u}_l\|^2\right) \tag{15}$$

where $\|\boldsymbol{u}_l\|^2=\boldsymbol{u}_l^H\boldsymbol{u}_l$ and σ^2 is the noise variance, assumed to be constant within the looks.

Under the hypothesis H_1, the data are modeled as a signal plus noise, hence, the pdf of the L independent looks $\{\boldsymbol{u}_l\}_{l=1}^{L}$ is given by

$$\begin{aligned}&f_{H_1}\left(\{\boldsymbol{u}_l\}_{l=1}^{L};\{\gamma_l\}_{l=1}^{L},\sigma^2,s\right)\\&\quad=\prod_{l=1}^{L}\frac{1}{\pi^M\sigma^{2M}}\exp\left(-\frac{1}{\sigma^2}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma_l\|^2\right).\end{aligned} \tag{16}$$

In (16) the case of neighboring pixels with different reflectivity values γ_l and same elevation s and noise variance σ^2 has been considered. The case of neighboring pixels having the same reflectivity can be easily obtained by setting $\gamma_1=\gamma_2=\cdots=\gamma_L=\gamma$. Both the cases are considered in the following.

A. Incoherent M-GLRT: Neighborhood of Targets With Same Elevation and Different Reflectivity

We consider first the case of a neighborhood of targets with the same elevation, but different reflectivity.

The GLRT test is defined as

$$\Lambda_1=\frac{\max\limits_{\{\gamma_l\}_{l=1}^{L},\sigma^2,s} f_{H_1}\left(\{\boldsymbol{u}_l\}_{l=1}^{L};\{\gamma_l\}_{l=1}^{L},\sigma^2,s\right)}{\max\limits_{\sigma^2} f_{H_0}\left(\{\boldsymbol{u}_l\}_{l=1}^{L};\sigma^2\right)}\underset{H_1}{\overset{H_0}{\gtrless}}\eta. \tag{17}$$

The value of σ^2 which maximizes the likelihood function under the null hypothesis (15) is given by

$$\begin{aligned}\hat{\sigma}_{H_0}^2&=\arg\max_{\sigma^2}\frac{1}{\pi^{LM}\sigma^{2LM}}\exp\left(-\sum_{l=1}^{L}\frac{1}{\sigma^2}\|\boldsymbol{u}_l\|^2\right)\\&=\arg\max_{\sigma^2}\left(-LM\ln(\pi)-LM\ln\left(\sigma_l^2\right)-\frac{1}{\sigma^2}\sum_{l=1}^{L}\|\boldsymbol{u}_l\|^2\right).\end{aligned} \tag{18}$$

To determine $\hat{\sigma}_{H_0}^2$, the derivative with respect to σ^2 is set equal to zero

$$\frac{\partial}{\partial\sigma^2}\left(-LM\ln(\pi)-LM\ln(\sigma^2)-\frac{1}{\sigma_l^2}\sum_{l=1}^{L}\|\boldsymbol{u}_l\|^2\right)=0.$$

Solving for σ^2, we have

$$\hat{\sigma}_{H_0}^2=\frac{1}{LM}\sum_{l=1}^{L}\|\boldsymbol{u}_l\|^2.$$

Consequently, the maximum of the likelihood function under H_0 is given by

$$\max_{\sigma^2} f_{H_0}\left(\{\boldsymbol{u}_l\}_{l=1}^{L};\sigma^2\right)=\frac{1}{\pi^{LM}\left(\frac{1}{LM}\sum_{l=1}^{L}\|\boldsymbol{u}_l\|^2\right)^{LM}}e^{-LM}. \tag{19}$$

In the hypothesis H_1, the likelihood function (16) has to be maximized with respect to σ^2, γ_l, with $l=1,\ldots,L$, and s. This maximization can be written as

$$\max_{\{\gamma_l\}_{l=1}^{L},\sigma^2,s}\frac{1}{\pi^{LM}\sigma^{2LM}}\exp\left(-\frac{1}{\sigma^2}\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma_l\|^2\right). \tag{20}$$

Let us consider first the maximization with respect to σ^2. We have

$$\begin{aligned}\hat{\sigma}_{H_1}^2&=\arg\max_{\sigma^2}\frac{1}{\pi^{LM}\sigma^{2LM}}\exp\left(-\sum_{l=1}^{L}\frac{1}{\sigma^2}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma_l\|^2\right)\\&=\arg\max_{\sigma^2}\left(-LM\ln\left(\sigma^2\right)-\frac{1}{\sigma^2}\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma_l\|^2\right).\end{aligned} \tag{21}$$

Once again, the derivative with respect to σ^2 is set equal to zero, with the aim to determine σ^2

$$-\frac{LM}{\sigma^2}+\frac{1}{(\sigma^2)^2}\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma_l\|^2=0.$$

Solving for σ^2 leads to

$$\hat{\sigma}^2_{H_1}=\frac{1}{LM}\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma_l\|^2. \tag{22}$$

Using (22), maximization (20) becomes

$$\max_{\{\gamma_l\}_{l=1}^{L},s}\frac{1}{\pi^{LM}\left(\frac{1}{LM}\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma_l\|^2\right)^{LM}}\exp(-LM). \tag{23}$$

The values of γ_l which maximize (23) are the ones which minimize the quadratic norms $\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma_l\|^2$, which are the least-squares solutions

$$\hat{\gamma}_l=\frac{\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l}{\boldsymbol{\varphi}(s)^H\boldsymbol{\varphi}(s)}=\frac{\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l}{M} \tag{24}$$

since $\boldsymbol{\varphi}(s)^H\boldsymbol{\varphi}(s)=M$.

Then, we have

$$\begin{aligned}&\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\hat{\gamma}_l\|^2\\&=\left(\boldsymbol{u}_l-\frac{1}{M}\boldsymbol{\varphi}(s)\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l\right)^H\left(\boldsymbol{u}_l-\frac{1}{M}\boldsymbol{\varphi}(s)\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l\right)\\&=\boldsymbol{u}_l^H\boldsymbol{u}_l-\frac{1}{M}\boldsymbol{u}_l^H\boldsymbol{\varphi}(s)\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l-\frac{1}{M}\boldsymbol{u}_l^H\boldsymbol{\varphi}(s)\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l\\&\quad+\frac{1}{M}\boldsymbol{u}_l^H\boldsymbol{\varphi}(s)\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l\\&=\boldsymbol{u}_l^H\boldsymbol{u}_l-\frac{1}{M}|\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l|^2.\end{aligned} \tag{25}$$

From (16), (22), and (25), it is obtained

$$\begin{aligned}&\max_{\{\gamma_l,\sigma_l^2\}_{l=1}^{L},s}belf_{H_1}\left(\{\boldsymbol{u}_l\}_{l=1}^{L};\{\gamma_l\}_{l=1}^{L},,\sigma^2,s\right)\\&=\max_{s}\frac{\exp(-LM)}{\pi^{LM}\left(\frac{1}{LM}\sum_{l=1}^{L}\left(\|\boldsymbol{u}_l\|^2-\frac{1}{M}|\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l|^2\right)\right)^{LM}}.\end{aligned} \tag{26}$$

Substituting (19) and (26) in (17), the following test is obtained:

$$\Lambda_1=\frac{\left(\sum_{l=1}^{L}\|\boldsymbol{u}_l\|^2\right)^{LM}}{\left(\sum_{l=1}^{L}\|\boldsymbol{u}_l\|^2-\max_s\sum_{l=1}^{L}\frac{1}{M}|\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l|^2\right)^{LM}}\underset{H_0}{\overset{H_1}{\gtrless}}\eta \tag{27}$$

$$\sum_{\ell=1}^{L}\|\boldsymbol{u}_\ell\|^2\underset{H_0}{\overset{H_1}{\gtrless}}\eta'\left(\sum_{\ell=1}^{L}\|\boldsymbol{u}_\ell\|^2-\max_s\sum_{l=1}^{L}\frac{1}{M}|\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l|^2\right) \tag{28}$$

leading to the incoherent M-GLRT test (11)

$$\frac{\max_s\sum_{l=1}^{L}\frac{1}{M}|\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l|^2}{\sum_{l=1}^{L}\boldsymbol{u}_l^H\boldsymbol{u}_l}\underset{H_0}{\overset{H_1}{\gtrless}}\frac{\eta'-1}{\eta'} \tag{29}$$

with $\eta'=\eta^{1/LM}$.

In (29), the incoherent superposition of the phase compensated looks in the considered neighborhood is performed.

B. Coherent M-GLRT: Neighborhood of Targets With Same Elevation and Same Reflectivity

We consider now the case of a neighborhood of targets with the same reflectivity, besides the same elevation.

Of course, the assumptions on the target elevation and reflectivity do not affect the pdf of the signal in the null hypothesis H_0 (absence of target), so that (19) remains unchanged.

Equation (20), instead, becomes

$$\max_{\gamma,\sigma^2,s}\frac{1}{\pi^{LM}\sigma^{2LM}}\exp\left(-\frac{1}{\sigma^2}\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma\|^2\right). \tag{30}$$

Following the same procedure described in Section V-A of the Appendix for the maximization with respect to σ^2, it can be easily obtained

$$\hat{\sigma}^2_{H_1}=\frac{1}{LM}\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma\|^2 \tag{31}$$

and the function to be maximized is

$$\max_{\gamma,s}\frac{1}{\pi^{LM}\left(\frac{1}{LM}\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma\|^2\right)^{LM}}\exp(-LM) \tag{32}$$

we note that

$$\begin{aligned}&\arg\max_{\gamma}\frac{1}{\left(\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma\|^2\right)^{LM}}\\&=\arg\min_{\gamma}\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma\|^2.\end{aligned} \tag{33}$$

To solve maximization (32), the derivative respect to γ has to be computed and set equal to zero

$$\begin{aligned}&\frac{\partial}{\partial\gamma}\left(\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\gamma\|^2\right)\\&=\frac{\partial}{\partial\gamma}\sum_{l=1}^{L}\left(\boldsymbol{u}_l^H\boldsymbol{u}_l-\gamma^H\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l-\boldsymbol{u}_l^H\boldsymbol{\varphi}(s)\gamma+\gamma^H\boldsymbol{\varphi}(s)^H\boldsymbol{\varphi}(s)\gamma\right)\\&=-2\sum_{l=1}^{L}\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l+2\left(\sum_{l=1}^{L}\boldsymbol{\varphi}(s)^H\boldsymbol{\varphi}(s)\right)\gamma=0.\end{aligned} \tag{34}$$

Then, the value of γ maximizing (32) is given by

$$\hat{\gamma}=\frac{\sum_{l=1}^{L}\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l}{\sum_{l=1}^{L}\boldsymbol{\varphi}(s)^H\boldsymbol{\varphi}(s)}=\frac{\sum_{l=1}^{L}\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l}{LM} \tag{35}$$

with $\sum_{l=1}^{L}\boldsymbol{\varphi}(s)^H\boldsymbol{\varphi}(s)=LM$. This value has to be substituted in (32). We have

$$\begin{aligned}&\sum_{l=1}^{L}\|\boldsymbol{u}_l-\boldsymbol{\varphi}(s)\hat{\gamma}\|^2\\&=\sum_{l=1}^{L}\left(\boldsymbol{u}_l^H\boldsymbol{u}_l-\hat{\gamma}^H\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l-\boldsymbol{u}_l^H\boldsymbol{\varphi}(s)\hat{\gamma}+\hat{\gamma}^H\boldsymbol{\varphi}(s)^H\boldsymbol{\varphi}(s)\hat{\gamma}\right)\\&=\sum_{l=1}^{L}\|\mathbf{u}_l\|^2-\frac{1}{LM}|\sum_{l=1}^{L}\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l|^2.\end{aligned} \tag{36}$$

By replacing (22) and (36), respectively, in the pdf under H_1, the following is obtained:

$$\max_{\gamma,s,\sigma^2} f_{H_1}\left(\{\boldsymbol{u}_l\}_{l=1}^{L};\gamma,s,\sigma^2\right) = \frac{e^{-LM}}{\pi^{LM}\left(\frac{1}{LM}\left(\sum_{l=1}^{L}\|\mathbf{u}_l\|^2-\max_s \frac{1}{LM}|\sum_{l=1}^{L}\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l|^2\right)\right)^{LM}}. \tag{37}$$

By replacing (19) and (37) in the GLRT test (17), after some mathematical manipulations, the following test is derived:

$$\Lambda_2 = \frac{\left(\sum_{l=1}^{L}\|\boldsymbol{u}_l\|^2\right)^{LM}}{\left(\sum_{l=1}^{L}\|\boldsymbol{u}_l\|^2-\max_s \frac{1}{LM}|\sum_{l=1}^{L}\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l|^2\right)^{LM}} \underset{H_0}{\overset{H_1}{\gtrless}} \eta \tag{38}$$

$$\sum_{\ell=1}^{L}\|\boldsymbol{u}_\ell\|^2 \underset{H_0}{\overset{H_1}{\gtrless}} \eta'\left(\sum_{\ell=1}^{L}\|\boldsymbol{u}_\ell\|^2-\max_s\frac{1}{LM}|\sum_{l=1}^{L}\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l|^2\right) \tag{39}$$

leading to the coherent M-GLRT test (12)

$$\frac{\max_s \frac{1}{LM}|\sum_{l=1}^{L}\boldsymbol{\varphi}(s)^H\boldsymbol{u}_l|^2}{\sum_{l=1}^{L}\boldsymbol{u}_l^H\boldsymbol{u}_l} \underset{H_0}{\overset{H_1}{\gtrless}} \frac{\eta'-1}{\eta'} \tag{40}$$

with $\eta' = \eta^{1/LM}$. In (40), the L looks are phase-compensated and coherently superimposed.

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